Where the Shortcut Sits

Status report 1: does masking the region of interest remove artifact shortcuts?

October 2026

Summary.

- Medical image classifiers often learn *artifacts* (rulers, hair, calipers, debris) instead of the disease. The usual defence is *ROI masking*: keep only the lesion or organ and delete the rest of the image.
- We found that masking works only when the artifact lies *outside* the region of interest (ROI). When the artifact lies *inside* the ROI, masking keeps it, removes everything else, and the model relies on the artifact *even more*.
- We showed this in four steps: (1) moving a ruler into a skin lesion step by step; (2) four checks of *why* it happens; (3) the same test in four other kinds of medical images; (4) real, unedited artifacts in four patient cohorts.
- With real artifacts, masking was worth much more when the artifact was outside the ROI than when it was inside, in all four cohorts (all four tests significant after correction for multiple testing).

1 The problem

A deep classifier learns whatever predicts the label in its training data, including things that are not the disease (Geirhos et al., 2020). This is called a *shortcut*. In medical images the shortcut is often an artifact that clinicians put there because they suspected disease: a ruler or skin marking next to a suspicious mole (Winkler et al., 2019, 2021), calipers on a thyroid nodule that was measured, or a chest drain in a patient with a collapsed lung (Jiménez-Sánchez et al., 2023). Because these artifacts appear more often in sick patients, a model can score well by detecting the artifact. It then fails on exactly the patients whose images look different, for example a melanoma photographed without a ruler.

The most common defence is to segment the lesion and give the model only that region; the rest of the image is deleted. This is called *region-of-interest (ROI) masking*. It is popular because it needs only a lesion mask, which many pipelines already compute. Its hidden assumption is that the artifact lies in the background. Rulers, hair, calipers and debris often lie on or across the lesion. Figure 1 shows what masking then does.

Research question. *Does the benefit of ROI masking depend on where the artifact sits relative to the ROI? If so, why, and does it hold for real artifacts and other kinds of medical images?*

2 How we test it

Two models Every experiment compares two models that use the same frozen image encoder (a pretrained foundation model such as DINOv2 (Oquab et al., 2024)) and the same simple classifier on top:

- the **standard model** sees the whole image;
- the **masked model** sees the image with everything outside the ROI replaced by a plain grey value, both in training and in testing.

We keep the encoder frozen so that the only difference between the two models is the input.

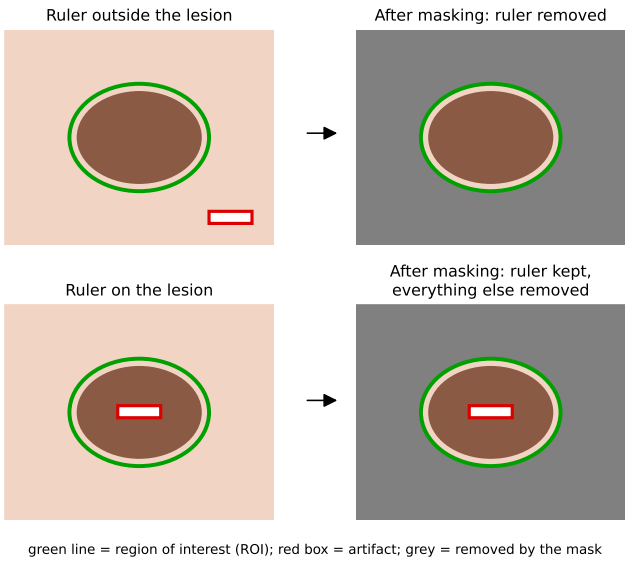


Figure 1: The idea. Top: an artifact outside the lesion is deleted by the mask. Bottom: an artifact on the lesion survives the mask, while everything that competed with it is deleted.

The trap To measure how much a model relies on the artifact, we build a "trap" (Figure 2). In the training data the artifact appears on 90% of the diseased images and 10% of the healthy ones, so it is a strong shortcut. We then test on three test sets:

- **correlated test**: same 90/10 pattern as in training. A shortcut model looks good here.
- **reversed test**: the pattern is flipped (10/90). A model that uses the artifact *fails* here.
- **clean test**: the artifact is unrelated to the disease. This shows how well the model knows the disease.

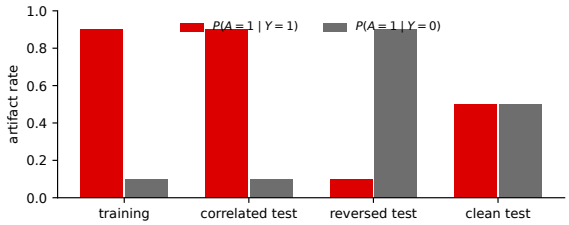


Figure 2: The trap. Bars show how often the artifact appears among diseased (red) and healthy (grey) images in training and in each test set.

The score We report AUROC (0.5 = guessing, 1.0 = perfect). Our main score is the AUROC on the *reversed* test: it is high only if the model looks at the disease and not at the artifact. We usually report the *difference* "masked minus standard". A positive number means masking helped; a negative number means it hurt. In square brackets we give the 95% confidence interval; if it does not include zero, the difference is significant.

Measuring fairly Two details turned out to matter.

- **Near-duplicate images.** Skin-lesion datasets contain several photos of the same lesion. If one copy is in training and another in testing, the test partly measures memory. We always split by lesion (and by image similarity), never by single images. With random splits, 93 near-duplicate pairs crossed the split and the measured shortcut was inflated by 10–18% (Table 1).
- **Honest error bars.** We train each model several times (different random seeds), but all runs are tested on the *same* test images. A common way of computing confidence intervals treats these runs

as if each had its own test set, which makes the intervals too narrow. We resample the seeds and the test images together, so the test set counts only once. Our intervals are about 1.5 times wider than the common method gives, and a few results that looked significant with the common method are not.

Before each main experiment we wrote down the design and what result would count as support, and stored it in the project repository; results are reported whichever way they went.

Table 1: Why we split by lesion: random image-level splits leak near-duplicates and inflate the shortcut (ruler experiment of Section 3, DINOv2). "Shortcut gap" = correlated minus reversed AUROC of the standard model.

Ruler position	Split	Near-duplicate pairs across the split	Shortcut gap	Masked − standard
outside the lesion	by lesion	0	0.241	+0.184
	random (mean of 5)	93	0.284	+0.223
on the lesion	by lesion	0	0.421	−0.129
	random (mean of 5)	93	0.465	−0.088

3 Step 1: moving a ruler into the lesion

What we did We first wanted to change *only* the location of the artifact. We took 2,437 dermoscopy images with manual lesion masks (ISIC 2018 (Codella et al., 2019); task: melanoma or not) and pasted the same small ruler into each image at 0, 25, 50, 75 or 100% overlap with the lesion (Figure 3). Everything else stays the same: same images, same ruler, same labels.

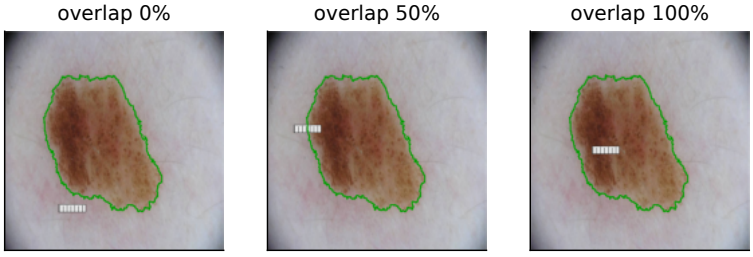


Figure 3: The same image with the ruler at 0, 50 and 100% overlap with the lesion (green: lesion mask). The pasted ruler is smaller than most real rulers so that it can lie fully inside the lesion; inside the lesion it covers about the same share of the lesion's area as real hair does (3.0% vs 3.3%).

What we found When the ruler lies outside the lesion, masking removes it and helps a lot (+0.18). As soon as the ruler overlaps the lesion, masking *hurts*: at full overlap the masked model is worse than the standard model (−0.13; Table 2, Figure 4). The clearest sign is on the correlated test: with the ruler on the lesion, the masked model scores *higher* than the standard model there (0.961 vs 0.937) but much lower on the reversed test (0.386 vs 0.515). It has become *more* dependent on the ruler.

Table 2: Ruler experiment (DINOv2, AUROC, mean of 3 seeds). The last row is the effect of masking on the reversed test.

	Ruler outside (0%)	Half on the lesion (50%)	Fully on the lesion (100%)
Standard model, reversed test	0.665	0.573	0.515
Masked model, reversed test	0.849	0.476	0.386
Standard model, correlated test	0.906	—	0.937
Masked model, correlated test	0.849	—	0.961
Masked − standard (reversed)	**+0.184** [+0.117, +0.250]	**−0.097** [−0.164, −0.030]	**−0.129** [−0.195, −0.064]

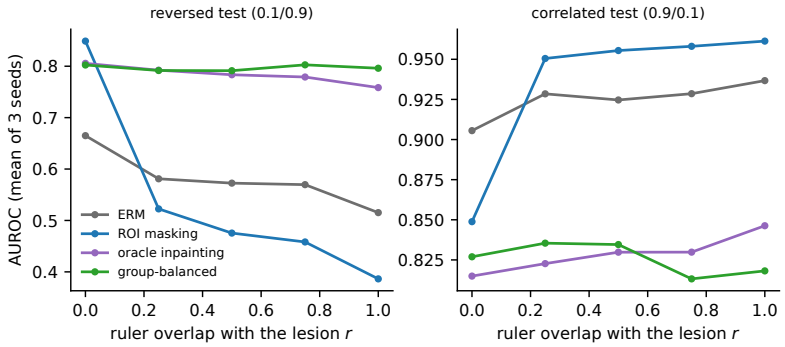


Figure 4: AUROC on the reversed test (left) and the correlated test (right) as the ruler moves into the lesion. Blue: masked model; grey: standard model. Green and purple: two methods that remove the ruler itself (they need to know where the ruler is), shown for comparison.

Is it a fluke of one model or one ruler? No. We repeated the experiment with a smaller image size, with a dermatology-specific encoder (DermLIP (Yan et al., 2025)), and with a ruler whose colour, ticks and size change from image to image. The pattern was the same every time (Table 3). We summarise it with one number, the *location effect*: the benefit of masking with the ruler outside minus its benefit with the ruler fully inside.

Table 3: The ruler result repeated with other settings (masked − standard, reversed test).

Setting	Ruler outside	Ruler fully inside	Location effect
DINOv2, large images (main result)	+0.184	−0.129	+0.313 [+0.258, +0.370]
DINOv2, small images	+0.361	−0.095	+0.456 [+0.390, +0.525]
DermLIP (dermatology encoder)	+0.161	−0.263	+0.424 [+0.335, +0.517]
DINOv2, randomised ruler	+0.146	−0.155	+0.302 [+0.248, +0.357]
The 1,314 images large enough to hold a long ruler fully inside the lesion:			
DINOv2, long thin ruler (200 × 10 px, like a real one)	+0.170	−0.054	+0.224 [+0.161, +0.285]
DINOv2, standard ruler, same images	+0.160	−0.102	+0.262 [+0.200, +0.325]

Does the ruler's size matter? The pasted ruler is smaller than most real rulers. We therefore repeated the experiment with a long, thin ruler of real proportions (39% of the image width, with millimetre ticks), on the 1,314 images whose lesion is large enough to hold it, and with the standard ruler on the same images for comparison (last two rows of Table 3). We wrote down the design and the success criteria before running it. The location effect holds (+0.224). With the long ruler fully on the lesion, masking loses all its benefit, but it is not clearly worse than the standard model (−0.054; the interval includes zero). The masked model again reacts to the ruler on the lesion about twice as strongly as the standard model (0.43 vs 0.21; see check 3 below).

4 Step 2: why does masking fail?

Step 1 showed *that* masking fails when the ruler is inside. There were three possible reasons: (a) masking simply hides useful parts of the image; (b) the background contains disease information that masking throws away; (c) the part of the ruler inside the lesion survives, and because masking deletes everything that competed with it, the model relies on it more. We designed one check for each (Table 4).

Explanation. Masking deletes everything outside the ROI. If the artifact is outside, the shortcut is gone. If the artifact is inside, it survives, and the other image information that used to compete with it is gone, so the artifact becomes a bigger part of what the model can use. The model relies on it more, and fails when the artifact is missing or misleading.

Table 4: Four checks of why masking fails (ruler experiment, DINOv2).

Check	What we changed	Result	What it means
1. Useless ruler	Ruler at the same places, but *unrelated* to the diagnosis	Masking does not hurt at any position (all intervals include 0; e.g. +0.039 at 100%)	Hiding parts of the image is not the problem (rules out a)
2. Bigger mask	Grow the mask around the lesion so that more of a half-covered ruler stays visible (50% → 95%)	Harm gets *worse*: −0.097 → −0.153; when the ruler is already fully visible, a bigger mask changes nothing (−0.129 vs −0.138)	Harm follows how much of the ruler is visible, not how much context is lost (rules out b)
3. Same-image test	Add or remove the ruler on the *same* test image and see how much the prediction changes	Masked model's prediction changes about 2 times as much as the standard model's (0.49 vs 0.24)	The masked model leans more on the ruler (supports c)
4. Lesion size	Split test images into small, medium and large lesions	Harm with the ruler inside: small −0.36, medium −0.16, large +0.06 (not significant)	Harm is largest where the ruler is big compared to the lesion

5 Step 3: is it only skin images?

What we did We repeated the moving-artifact experiment in four other kinds of medical images, each with its own typical artifact and its own ROI (Figure 5): a sonographer's caliper on **thyroid ultrasound** (TN3K, malignant or benign nodule (Gong et al., 2021)) and on **ovarian ultrasound** (MMOTU, tumour with or without a solid part (Zhao et al., 2022)); a debris blob on **capsule endoscopy** (SEE-AI, erosion or polyp-like lesion (Yokote et al., 2023)); and a tube on **chest X-rays** (NIH ChestX-ray14, pneumothorax (Wang et al., 2017); ROI = the lungs). We used only images without a real artifact, so the pasted one was the only one. We also tested a second encoder where possible (MedSigLIP, a medical image model; RAD-DINO, a chest X-ray model (Pérez-García et al., 2025)).

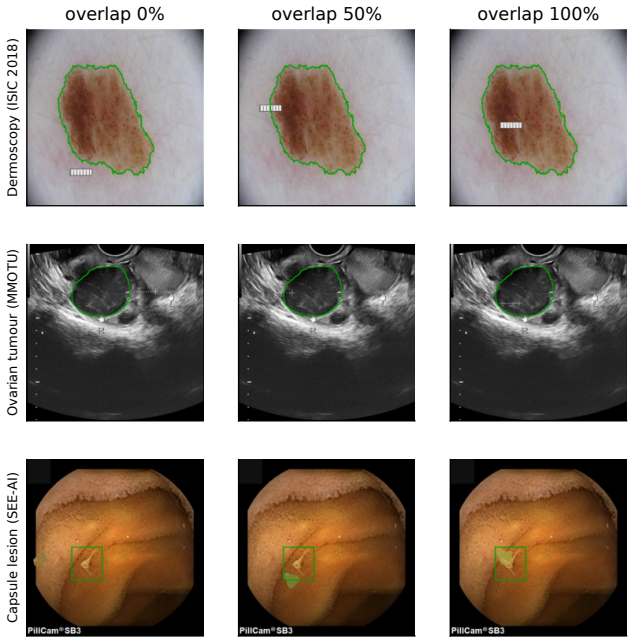


Figure 5: The pasted artifact at 0, 50 and 100% overlap in three modalities (green: ROI; for capsule images the ROI is the expert's box around the lesion).

What we found The location effect was positive in 10 of the 11 experiments of Steps 1 and 3 (Table 5, Figure 6). Masking helped clearly when the artifact was outside and lost most or all of its benefit when the artifact was inside. Whether masking actually *hurt* at full overlap depended on the setting: it hurt in 7 of 11 experiments. In thyroid ultrasound, for example, the benefit dropped from +0.40 to about zero.

Table 5: Moving-artifact experiments in other modalities (masked − standard, reversed test).

Modality and artifact	Encoder	Artifact outside	Artifact fully inside	Location effect
Thyroid ultrasound, caliper	DINOv2	+0.402	−0.030	+0.433 [+0.362, +0.507]
	MedSigLIP	+0.514	+0.044	+0.470 [+0.388, +0.556]
Ovarian ultrasound, caliper	DINOv2	+0.041	−0.197	+0.238 [+0.144, +0.340]
Capsule endoscopy, debris	DINOv2	+0.116	−0.112	+0.228 [+0.144, +0.324]
	MedSigLIP	+0.374	+0.216	+0.158 [+0.012, +0.314]
Chest X-ray, tube	DINOv2	+0.254	−0.083	+0.337 [+0.312, +0.364]
	RAD-DINO	−0.002	+0.111	−0.113 [−0.129, −0.097]

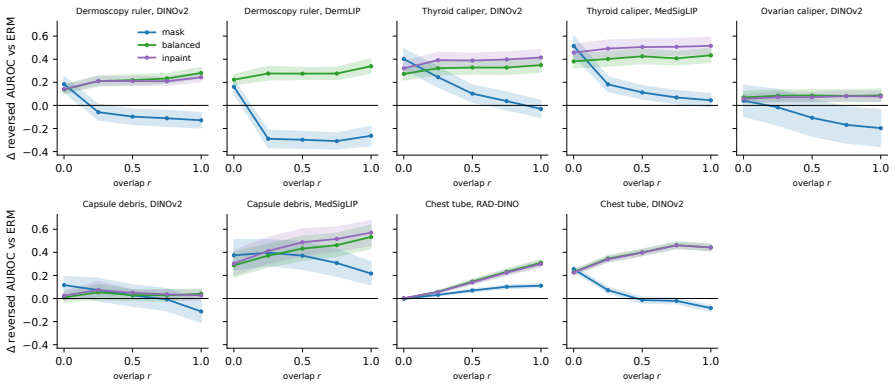


Figure 6: Effect on the reversed test (vs the standard model) as the artifact moves into the ROI, in every moving-artifact experiment. Blue: masking; green and purple: methods that remove the artifact itself. The blue line falls in every panel except the RAD-DINO chest tube.

The one exception With the chest X-ray encoder RAD-DINO, the standard model simply ignored a tube outside the lungs (reversed AUROC 0.913, the same as on the clean test, 0.915). There was nothing for masking to remove outside, so the comparison does not apply. With DINOv2, which did use the outside tube, the usual pattern appeared.

6 Step 4: real artifacts

Pasted artifacts let us control location exactly, but they are not the artifacts clinicians produce. So we next asked whether the same thing happens with *real, unedited* artifacts.

The design A real artifact cannot be moved, so we compare two groups of images: those whose artifact lies mostly inside the ROI (**Trap A**) and those whose artifact lies outside it (**Trap B**). Both traps share the *same* artifact-free images (Figure 7), so the main designed difference between them is where the artifact sits. (We first tried comparing in-ROI against out-of-ROI images directly, without an artifact-free group. That design cannot work: masking deletes exactly the out-of-ROI artifact that defines one side of the comparison, so it shows "harm" whatever the truth.) Each trap then gets the 90/10 training set and the three test sets of Figure 2. The **crossover** is the benefit of masking in Trap B minus its benefit in Trap A; a positive crossover means masking is worth more when the artifact is outside.

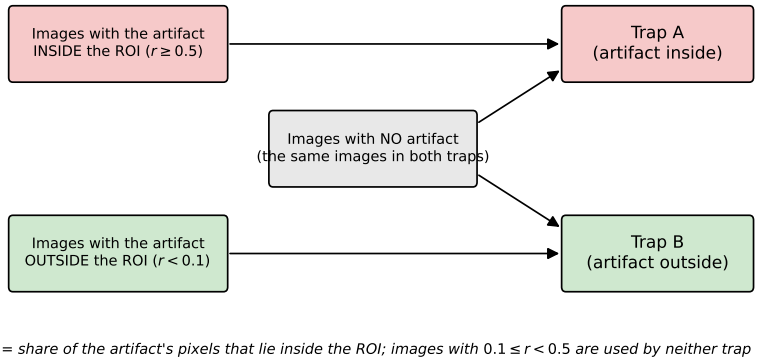


Figure 7: Two traps built from real images. Both share one group of artifact-free images.

Cohorts Table 6. Artifacts were located automatically: published hair masks for skin images (Wegley et al., 2026), a rule-based caliper detector for ultrasound, and a small detector trained on expert debris masks for capsule images. We checked them by eye: the caliper detector was right on 38 of 40 checked images. Detection errors cannot create a crossover: they weaken the shortcut in both traps equally.

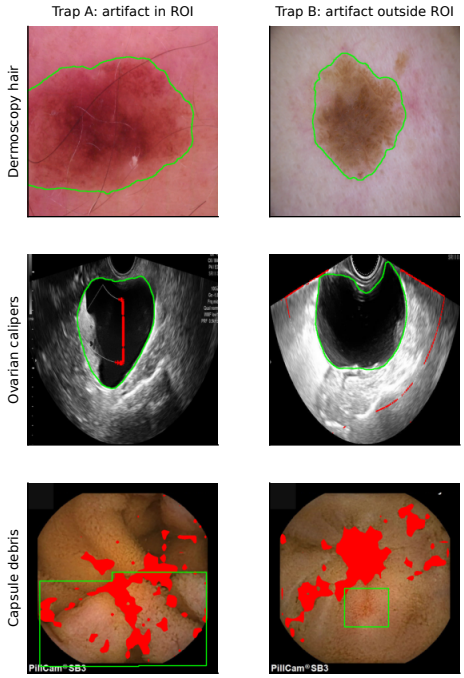


Figure 8: Real artifacts inside (Trap A) and outside (Trap B) the ROI. Green: ROI; red: detected artifact. (Thyroid images are not shown because the data licence is unclear; hair masks are not drawn for licence reasons.)

Table 6: The four real-artifact cohorts (number of images, diseased / not diseased).

Cohort	Task	Artifact	No artifact	Trap A (inside)	Trap B (outside)
Skin lesions, ISIC 2019 (3 hospitals)	melanoma vs other	hair	157 / 706	942 / 2,655	969 / 4,412
Thyroid ultrasound, TN3K	malignant vs benign	calipers	749 / 1,052	292 / 858	99 / 227
Ovarian ultrasound, MMOTU	solid part vs cystic	calipers	178 / 170	276 / 341	103 / 91
Capsule endoscopy, SEE-AI	erosion vs polyp-like	debris, bubbles	343 / 164	412 / 404	1,009 / 231

What we found The crossover was positive in all four cohorts with DINOv2 (Table 7). These four tests were planned in advance, and all four stay significant after Holm correction for testing four cohorts (adjusted $p = 0.0004$ each). The result also held with two other encoders, including a convolutional network (ConvNeXt), so it is not specific to one type of model (Figure 9).

Table 7: Real artifacts: effect of masking on the reversed test (masked − standard) in each trap, and the crossover.

Cohort	Encoder	Trap A (inside)	Trap B (outside)	Crossover (B − A)
Skin, hair	DINOv2	−0.049	+0.098	+0.147 [+0.111, +0.185]
Thyroid, calipers	DINOv2	+0.109	+0.348	+0.239 [+0.196, +0.280]
Ovary, calipers	DINOv2	+0.132	+0.302	+0.170 [+0.087, +0.252]
Capsule, debris	DINOv2	+0.084	+0.461	+0.377 [+0.327, +0.426]
Thyroid, calipers	MedSigLIP	+0.053	+0.248	+0.195 [+0.153, +0.238]
Ovary, calipers	MedSigLIP	+0.002	+0.180	+0.178 [+0.109, +0.248]
Capsule, debris	MedSigLIP	+0.117	+0.452	+0.335 [+0.286, +0.383]
Thyroid, calipers	ConvNeXt	−0.004	+0.271	+0.275 [+0.231, +0.320]
Capsule, debris	ConvNeXt	+0.084	+0.403	+0.320 [+0.268, +0.372]
Skin, hair	DermLIP	−0.104	−0.090	+0.014 [−0.037, +0.063]

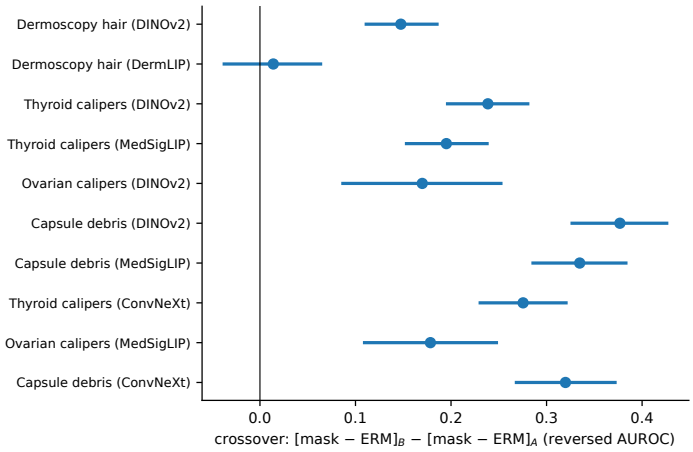


Figure 9: The crossover for every cohort and encoder (dots: estimate; lines: 95% interval). Right of zero = masking is worth more when the artifact is outside the ROI.

How to read it For skin hair, masking actually *hurt* when the hair lay on the lesion. For calipers and debris, masking still helped a little in Trap A, because it removes other on-screen marks and clutter, but a large part of the shortcut stayed. In thyroid Trap A, the masked model still scored 0.908 on the correlated test but only 0.398 on the reversed one; in Trap B masking closed this gap (0.785 vs 0.752). The general rule is therefore: *masking is worth much less when the artifact is inside the ROI.* Whether it actually hurts depends on how much useful information masking also removes.

7 Other things we learned

- **Methods that know where the artifact is fix both traps.** Reweighting the training images so that the artifact no longer predicts the label, or retraining the last layer on balanced data (DFR (Kirichenko et al., 2023)), helped in both traps of every cohort, wherever the artifact was (Table 8). The catch is that they need an artifact label for every training image, which clinical datasets rarely have; masking needs only the ROI.

- **Painting out the artifact can backfire.** In capsule images, painting over the detected debris *lowered* the score (-0.17 in Trap A), because erosions contain yellow-white material that looks like debris. Deleting artifact pixels is only safe when the artifact does not look like the disease.
- **The hair shortcut is not only in the hair pixels.** Hair goes together with patient age, sex and body site. Painting out the hair pixels closed less than half of the standard model's shortcut gap (47% in Trap A, 44% in Trap B); the rest comes from things that go with hair. Masking cannot remove such a shortcut.
- **One encoder did not follow the rule.** With DermLIP, masking hurt in *both* hair traps and there was no crossover: this encoder needs the skin around the lesion, so any mask costs disease information.
- **Chest X-rays are a boundary case.** Real chest drains lie inside the lungs and are an extreme shortcut (standard model: reversed AUROC 0.19 against 0.89 on the clean test); lung masking barely helped (0.23), while DFR reached 0.67. With real devices (catheters in the lungs vs breathing tubes outside), RAD-DINO showed no crossover, while MedSigLIP showed a positive one in all four findings tested. The device types also differ in which other devices are present, so we do not count chest X-rays as a confirmation.

Table 8: Masking against methods that use artifact labels (reversed test, change vs the standard model, DINOv2).

	Skin, hair		Thyroid, calipers		Capsule, debris	
Method	Trap A	Trap B	Trap A	Trap B	Trap A	Trap B
ROI masking (needs ROI only)	−0.049	+0.098	+0.109	+0.348	+0.084	+0.461
Balanced reweighting (needs labels)	+0.213	+0.204	+0.388	+0.254	+0.179	+0.244
DFR (needs labels)	+0.196	+0.196	+0.463	+0.312	+0.234	+0.287
Painting out the artifact (needs its mask)	+0.098	+0.100	+0.102	+0.039	−0.172	−0.115

8 Where we are

Table 9: What the work so far shows.

Question	Answer	Evidence
Does masking depend on where the artifact sits?	Yes: it helps when the artifact is outside the ROI and loses its benefit (often turns harmful) when it is inside	Steps 1 and 3: 10 of 11 experiments
Why?	The artifact inside survives, its competitors are deleted, and the model relies on it more	Step 2: four checks
Does it hold for real artifacts?	Yes, in all four cohorts and with three encoder types	Step 4: four planned tests, all significant
Exceptions	An encoder that needs the lesion's surroundings (DermLIP); an encoder that ignores the outside artifact (RAD-DINO)	Steps 3 and 4

Questions for discussion

1. Is the reversed-test AUROC acceptable as the main score, or should a clinical measure (sensitivity at a fixed specificity) be added from the start?

2. Should the rule be stated as "masking is worth less inside the ROI" (true in all real cohorts) rather than "masking hurts inside the ROI" (true for skin hair and in 7 of 11 moving-artifact experiments)?
3. The artifact masks were checked by eye by a non-clinician. Should a clinician check a sample?
4. How should chest X-rays be presented: as a boundary case, or left out of the main text?

References

Noel Codella, Veronica Rotemberg, Philipp Tschandl, et al. Skin lesion analysis toward melanoma detection 2018: A challenge hosted by the international skin imaging collaboration (ISIC). *arXiv:1902.03368*, 2019.

Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A Wichmann. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2:665–673, 2020.

Haifan Gong, Guanqi Chen, Ranran Wang, Xiaoliang Xie, Mingzhi Mao, Yizhou Yu, Fei Chen, and Guanbin Li. Multi-task learning for thyroid nodule segmentation with thyroid region prior. In *IEEE International Symposium on Biomedical Imaging (ISBI)*, 2021. doi: 10.1109/ISBI48211.2021.9434087.

Amelia Jiménez-Sánchez, Dovile Juodelyte, Bethany Chamberlain, and Veronika Cheplygina. Detecting shortcuts in medical images – a case study in chest x-rays. In *IEEE ISBI*, 2023.

Polina Kirichenko, Pavel Izmailov, and Andrew Gordon Wilson. Last layer re-training is sufficient for robustness to spurious correlations. In *ICLR*, 2023.

Maxime Oquab, Timothée Darcet, Théo Moutakanni, et al. DINOv2: Learning robust visual features without supervision. *TMLR*, 2024.

Fernando Pérez-García, Harshita Sharma, Sam Bond-Taylor, et al. Exploring scalable medical image encoders beyond text supervision. *Nature Machine Intelligence*, 2025.

Xiaosong Wang, Yifan Peng, Le Lu, Zhiyong Lu, Mohammadhadi Bagheri, and Ronald M Summers. ChestX-ray8: Hospital-scale chest x-ray database and benchmarks. In *CVPR*, 2017.

Victoria Wegley et al. Dermatology skin lesion image analysis: artifact annotation masks for ISIC 2019. Scholars' Mine, Missouri S&T, doi:10.71674/man1-qa33, 2026.

Julia K Winkler, Christine Fink, Ferdinand Toberer, et al. Association between surgical skin markings in dermoscopic images and diagnostic performance of a deep learning convolutional neural network for melanoma recognition. *JAMA Dermatology*, 155(10):1135–1141, 2019.

Julia K Winkler, Katharina Sies, Christine Fink, et al. Association between different scale bars in dermoscopic images and diagnostic performance of a market-approved deep learning convolutional neural network for melanoma recognition. *European Journal of Cancer*, 145:146–154, 2021.

Siyuan Yan et al. Derm1M: A million-scale vision-language dataset aligned with clinical ontology knowledge for dermatology. In *ICCV*, 2025.

Akihito Yokote et al. The SEE-AI project dataset: capsule endoscopy images with lesion bounding boxes. Kaggle dataset capsuleyolo/kyucapsule, CC BY 4.0, 2023.

Qi Zhao, Shuchang Lyu, Wenpei Bai, Linghan Cai, Binghao Liu, Meijing Wu, Xiubo Sang, Min Yang, and Lijiang Chen. A multi-modality ovarian tumor ultrasound image dataset for unsupervised cross-domain semantic segmentation. *arXiv preprint arXiv:2207.06799*, 2022.